



# Knowledge Institute of Technology

Centre for Career Development & Training (CDT)

Summative Assessment I (Pen & Paper Mode)

Smartcliff & Silicon (SM – 4 & 5 and SI – 4 & 5)

Date: 29.06.2026 Duration: 1 Hr. 15 Mins.

Max.Marks: 100 Batch : 2027 / IV Year

Section – A (60 Minutes)

Programming Questions (80 Marks)

1. A timekeeper has a collection of clocks, each representing a specific hour. Your task is to calculate the total hours represented by the clocks and find the clock showing the highest hour. (10 Marks)

Sample Input:

4

3 7 5 9

Sample Output:

24 9

2. A streaming service allows users to maintain a personal watchlist. Since users may intentionally add the same movie multiple times, duplicate movie titles are allowed. Use a List<String> to manage the watchlist. (20 Marks)

Create a class MovieWatchList and implement the following methods.

```
void addMovie(String movie)
```

```
void removeMovie(String movie)
```

```
boolean containsMovie(String movie)
```

```
int movieCount()
```

```
List<String> getMovies()
```

Sample Input

```
addMovie("Avatar");
```

```
addMovie("Inception");
```

```
addMovie("Avatar");
```

```
print(movieCount());  
print(getMovies());  
print(containsMovie("Titanic"));
```

Sample Output

3

[Avatar, Inception, Avatar]

false

Sample Input 2

```
addMovie("Avatar");
```

```
addMovie("Titanic");
```

```
addMovie("Joker");
```

```
removeMovie("Titanic");
```

```
print(containsMovie("Titanic"));
```

```
print(movieCount());
```

```
print(getMovies());
```

Sample Output 2

false

2

[Avatar, Joker]

3. A software company maintains information about its employees and managers. Every manager is an employee, but in addition to the common employee details, a manager is also responsible for leading a team.

Design a Java application to model this scenario using inheritance. The program should demonstrate constructor chaining using super() and method overriding to display role-specific information. (30 Marks)

Task

Create the Base Class named Employee with the following data members:

- employeeId (int)
- employeeName (String)
- department (String)

Constructor

Employee(int employeeId, String employeeName, String department)

Method

public void displayDetails() that displays

- Employee ID
- Employee Name
- Department
- Role as Employee

Create the Derived Class named Manager that inherits from the Employee class.

Add the following data member:

- teamSize (int)

Constructor

Manager(int employeeId, String employeeName, String department, int teamSize)

Method Overriding

@Override

public void displayDetails() that displays:

- Employee ID
- Employee Name
- Department
- Team Size
- Role as Manager

Constraints

- Employee ID > 0
- Team Size ≥ 1
- Employee Name and Department are non-empty strings.

Sample Input

101

John

Information Technology

201

Alice

Human Resources

12

Sample Output

Employee Details

Employee ID : 101

Employee Name : John

Department : Information Technology

Role : Employee

Manager Details

Employee ID : 201

Employee Name : Alice

Department : Human Resources

Team Size : 12

Role : Manager

**4. Given a singly linked list, perform the following operations on it in order: (40 Marks)**

- a. Insert a new node with a given value immediately after a specified position.
- b. Delete the node at a specified position.
- c. Reverse the entire linked list.

Return the final linked list after all three operations have been applied.

Input & explanation:

3 → number of initial nodes

1 2 4 → the 3 initial values

3 2 → insert value 3 after position 2

2 → delete the node at position 2 (from the updated 4-node list)

Output:

4 3 1 → final list after insert, delete, and reverse

(Contd. to Section B)

**Section – B (15 Minutes)**

**Aptitude Questions ( 10 \* 2 = 20)**

- 1) A car travels from City P to City Q at a speed of 60 km/hr and returns from Q to P at a speed of 40 km/hr. What is the average speed of the car for the entire journey?  
A) 45 km/hr                      B) 48 km/hr                      C) 50 km/hr                      D) 52 km/hr                      E) 55 km/hr
- 2) A train 240 metres long is running at a speed of 72 km/hr. In how many seconds will it cross a platform 360 metres long?  
A) 20 seconds      B) 24 seconds      C) 28 seconds      D) 30 seconds      E) 36 seconds
- 3) A boat covers 32 km downstream in 4 hours and the same distance upstream in 8 hours. What is the speed of the boat in still water?  
A) 3 km/hr                      B) 4 km/hr                      C) 5 km/hr                      D) 6 km/hr                      E) 8 km/hr
- 4) A sum of money amounts to ₹9,440 in 3 years and ₹10,400 in 5 years at simple interest. What is the principal sum?  
A) ₹7,800                      B) ₹7,900                      C) ₹8,000                      D) ₹8,060                      E) ₹8,180
- 5) Find the compound interest on ₹15,000 for 2 years at 10% per annum, compounded annually.  
A) ₹3,000                      B) ₹3,100                      C) ₹3,150                      D) ₹3,200                      E) ₹3,300
- 6) In an examination, a student scored 35% marks and failed by 12 marks. Another student scored 50% marks and obtained 18 marks more than the minimum passing marks. What is the maximum mark of the examination?  
A) 180                      B) 190                      C) 195                      D) 200                      E) 210
- 7) A shopkeeper marks an article 40% above its cost price and then allows a discount of 15% on the marked price. What is his overall profit percentage?  
A) 16%                      B) 18%                      C) 19%                      D) 21%                      E) 24%
- 8) Five friends — Manoj, Neha, Om, Priya, and Rahul — are ranked based on their salaries. Neha earns more than Om but less than Priya. Rahul earns more than Priya. Manoj earns less than Om. Who earns the highest salary?  
A) Manoj                      B) Neha                      C) Om                      D) Priya                      E) Rahul
- 9) Six people — J, K, L, M, N, and O — sit in a row facing north. K sits third from the left end. M sits immediately to the right of K. L sits at one of the extreme ends. N sits second from the right end. J sits immediately to the left of O. Who sits at the left extreme end?  
A) J                      B) K                      C) L                      D) N                      E) O
- 10) In a class of 40 students, Aman ranks 14th from the top. Sonia ranks lower than Aman, with exactly 5 students positioned between them. How many students rank below Sonia in the class?  
A) 18                      B) 19                      C) 20                      D) 21                      E) 22

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